

USING SESSIONS MANAGER IN AWS SYSTEMS MANAGER TO CONNECT TO EC2 INSTANCE.

1) CREATE A ROLE AND ATTACH THE AmazonSSMManagedInstanceCore POLICY (TO ENABLE SYSTEMS MANAGER CONNECT TO YOUR EC2 INSTANCE. WITHOUT THIS ROLE, THE INSTANCE WOULD NOT SHOW UP AS "MANAGED" IN SYSTEMS MANAGER) AND THE AmazonSSMFullAccess POLICY TO ALLOW YOUR ACCOUNT (ADMIN) TO CONNECT AND MANAGE RESOURCES IN SYSTEMS MANAGER.

CREATE THE ROLE

- a) type iam in your console
- b) under ACCESS MANAGEMENT click on 'Roles' locate 'Create role' and click on it.
- c) under TRUSTED ENTITY TYPE choose 'AWS service'.
- d) under USE CASE select EC2 and click next.
- e) under PERMISSIONS POLICY search for AmazonSSMManagedInstanceCore AND CLICK ON THE CHECK BOX and search for AmazonSSMFullAccess and click on the CHECK BOX and click next.
- f) under ROLE NAME give your role name "webserver_role" and click on create role.

2) CREATE AN EC2 INSTANCE AND ATTACH THE ROLE FROM ABOVE.

- a) Click on launch EC2 instance and give your EC2 instance the name, 'mywebserver'.
- b) select the any Amazon AMI for free tier.
- c) under the INSTANCE TYPE select t2 or t3 micro.
- d) under KEY PAIR select "proceed without key pair (not recommended)" because we do not want to SSH into this EC2 instance we want to use session manager.
- e) under NETWORK SETTINGS click on EDIT and under "inbound security group rules" remove the default security group rule and click on add security group. under TYPE click on "HTTP" and under SOURCE TYPE click on "Anywhere". under description write "mywebserverSG".
- f) under ADVANCED DETAILS click on "IAM instance profile" and attach the "webserver_role" role we created initially to enable session manager to connect to our EC2 instance.

g) under USER DATA paste the bash script below.

```
#!/bin/bash

# Install nginx
yum install -y nginx

systemctl start nginx # Start nginx web server
systemctl enable nginx # Configure nginx to auto-run when the Operating System starts

# Create a simple index.html file
echo "<html>
<head>
    <title>My Nginx Server</title>
</head>
<body>
    <h1>Hello Hardworking Nigerian Youths</h1>
</body>
</html>" > /usr/share/nginx/html/index.html

# Restart nginx to load the new index.html
systemctl restart nginx
```

We are using Nginx to deliver a static HTML file. It can also be used for CSS, JavaScript, images, or videos to users' browsers. Nginx is pronounced as "engine-x" is a high-performance web server and reverse proxy. Nginx will show our HTML page because it is serving from index.html using the /usr/share/nginx/html/index.html folder in the Nginx application. Nginx is a fast, efficient tool for delivering web content, protecting your apps, and managing traffic. But for this purpose, it will serve as a web content.

h) click on launch instance.

3) ACCESS SESSION MANAGER (PERFORM WITH YOUR INSTRUCTOR)

a) click on the check box of the created 'mywebserver' EC2 instance and at the top of the page click on CONNECT.

b) locate session manager and click on CONNECT.

c) in the terminal, run the command below to check if Nginx is installed.

```
nginx -v
```

d) run the command below to check the user data logs to confirm if our user data was successfully installed in the 'mywebserver' EC2 instance.

```
sudo cat /var/log/cloud-init-output.log
```

e) run the commands below to test the web service locally. you should see our html structure.

this

```
curl http://127.0.0.1
```

or this

```
curl http://localhost
```

f) run the command below to start the nginx engine on the 'mywebserver' EC2 instance.

```
sudo service nginx status
```

g) locate the 'mywebserver' EC2 instance and copy the Public IPv4 address, you should see something like '13.53.71.103' and paste in a new web browser and you should see the page hosted and running.

h) run the command below to stop the nginx engine on the 'mywebserver' EC2 instance. Try refreshing the browser where your page is and confirm your result.

```
sudo service nginx stop
```

h) stop your EC2 instance.